RoundBot – Assembly Guide

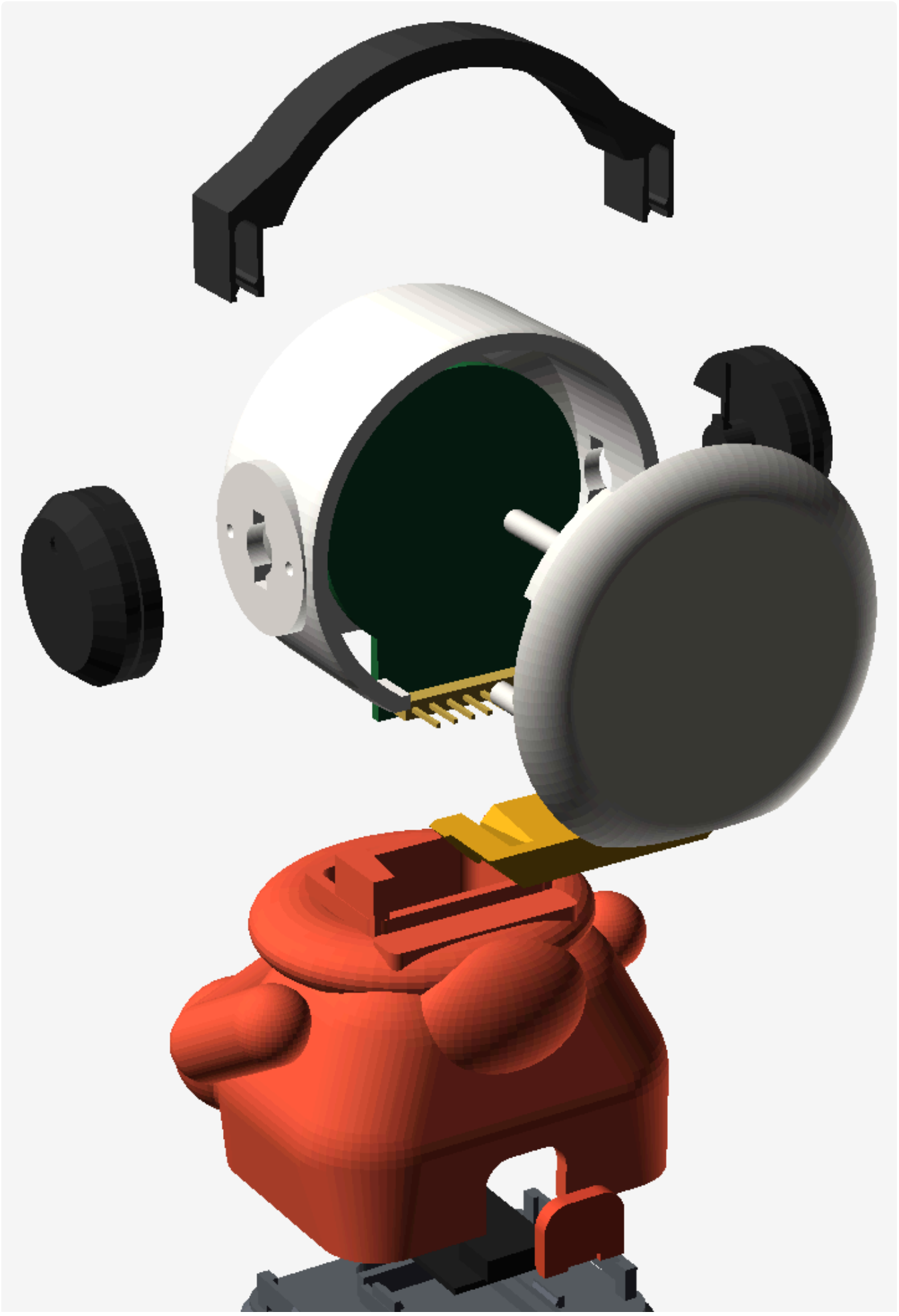
Nov 8, 2026 · @michael

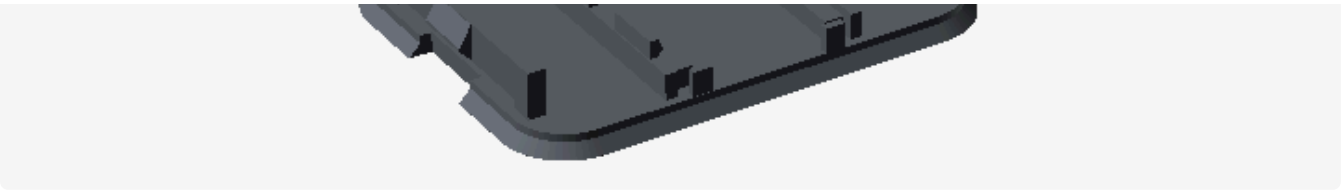
Overview

RoundBot is a 92 mm tall desk companion. Its face is the 1.28″ GC9A01 round display: the white bezel shows only the 33 mm active area, so the eyes you draw in firmware are the face. An ESP32-S3 SuperMini sits in the base, and its USB-C port opens through a notch low on the back, behind a removable cover.

Version 8 needs no glue and no screws. A wedge key locks the head onto the neck, and the back cap snaps in behind it. The side pieces twist-lock into the head on a sprung peg, and the band snaps over a bar in each cup. A bare template and twelve themes share one core, and their head accessories swap freely.







Core parts (every theme)

File	Colour	What it does
core/head_shell.stl	White	Face bezel, display pocket, keyhole sockets on both sides
core/head_cap.stl	White	Back of head; posts clamp the display, and it closes behind the neck key
core/neck_key.stl	Any	Wedge that slides in behind the display and locks the head onto the neck
core/base.stl	Any	Bottom plate that snaps into the body, with a snap-in cradle for the SuperMini
core/usb_cover.stl	Body colour	Friction plug that hides the USB-C notch

Themes

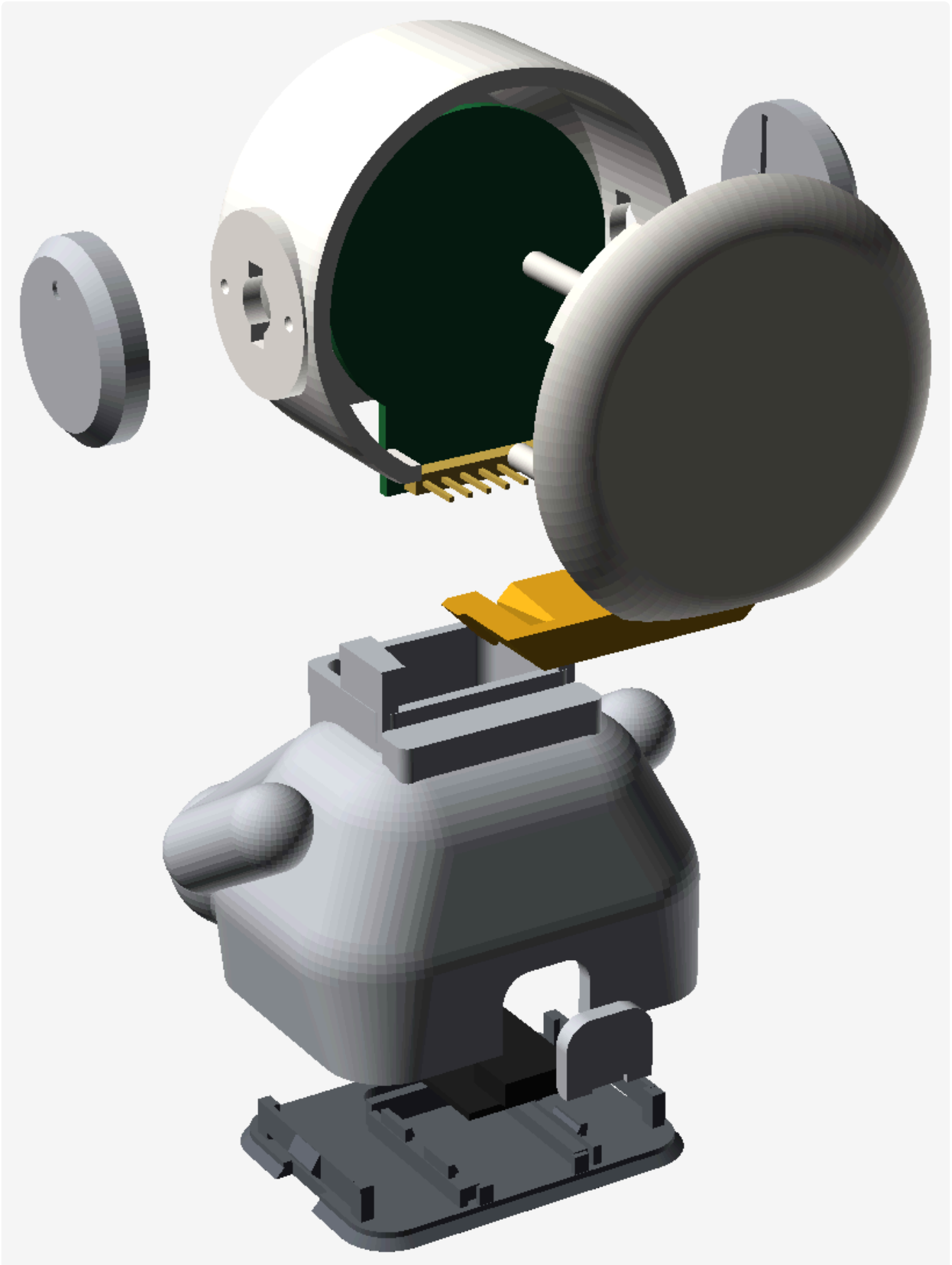
Print the core once, then one body and one set of head accessories. Colours are only suggestions.

Version	Body	Head accessories	Colour idea
Template (bare)	body_bare	plug x2	Any single colour
Skeleton	body_skeleton (rib relief, bone limbs)	plug x2	Bone white or glow-in-the-dark
DJ	body_hoodie	cup_left, cup_right, band_plain	Orange hoodie, black headphones
Gamer	body_hoodie	cup_mic (left), cup_right, band_plain	Blue hoodie, black headset
Cat	body_cat (hoodie with tail)	cup_left, cup_right, band_cat	Grey body, pink headset
Bear	body_hoodie	cup_left, cup_right, band_bear	Brown

Version	Body	Head accessories	Colour idea
Bunny	body_bare	cup_left, cup_right, band_bunny	White, pink cups
Bug	body_hoodie	cup_left, cup_right, band_bug	Yellow body, black headset
Devil	body_cat (the tail again)	cup_left, cup_right, band_devil	Red body, dark red horns
King	body_hoodie	cup_left, cup_right, band_crown	Purple body, gold crown
Dino	body_cat (the tail again)	cup_left, cup_right, band_dino	Green body, orange plates
Unicorn	body_bare	cup_left, cup_right, band_unicorn	White body, lilac headset
Astronaut	body_astro (suit collar, backpack)	pod_left, pod_right (antenna)	White body, silver pods

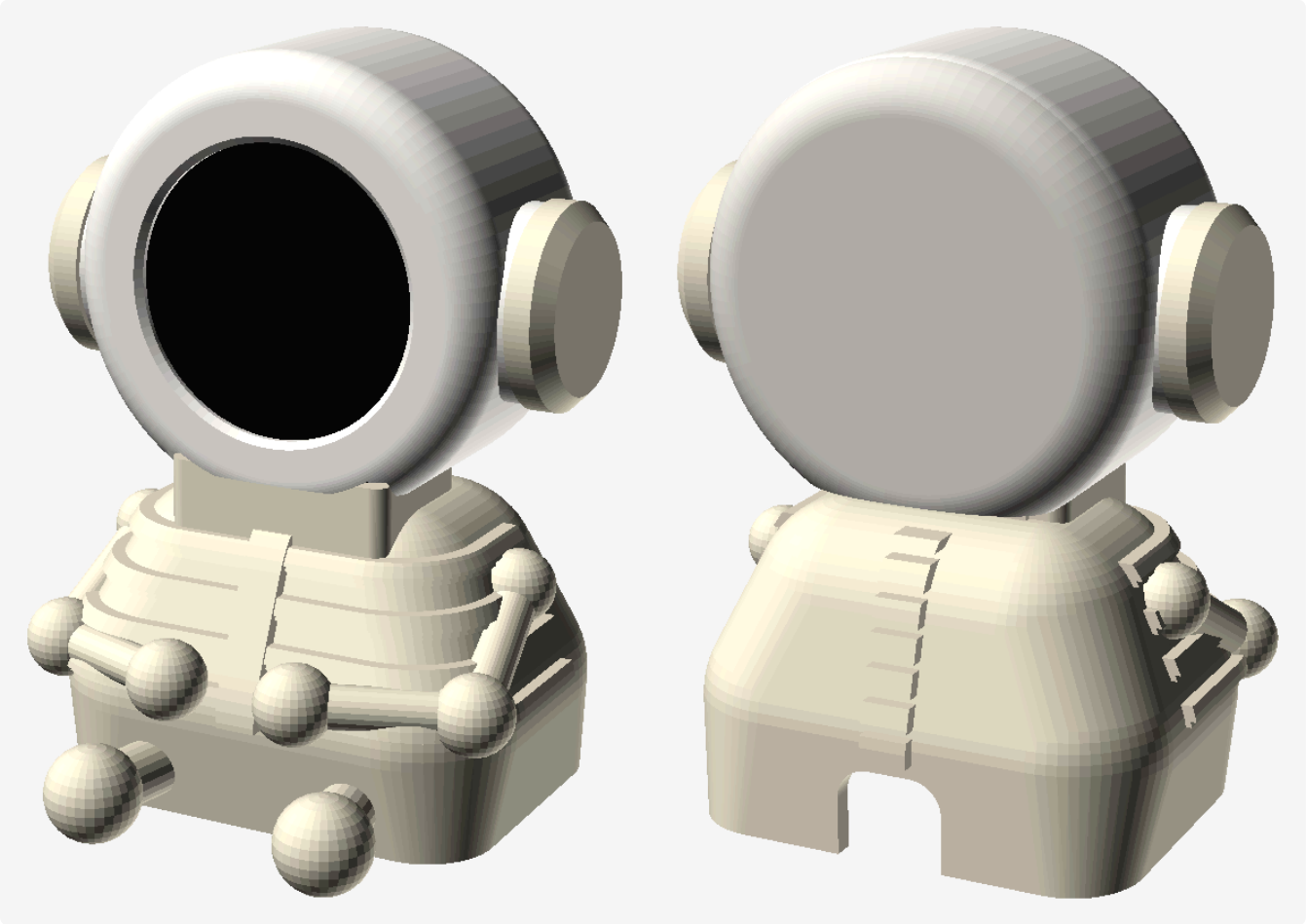
Template (bare kit)

The template is the robot with nothing themed on it: the five core parts, body_bare, and two plugs that close the keyholes on the head. One plug STL fits both sides. It works as a finished robot and as the base for your own theme.



Skeleton

The skeleton body has raised ribs, a sternum and a spine on the torso, and thin bones with knobby joints for arms and legs. The electronics stay enclosed. Draw skull eyes on the display to finish it.



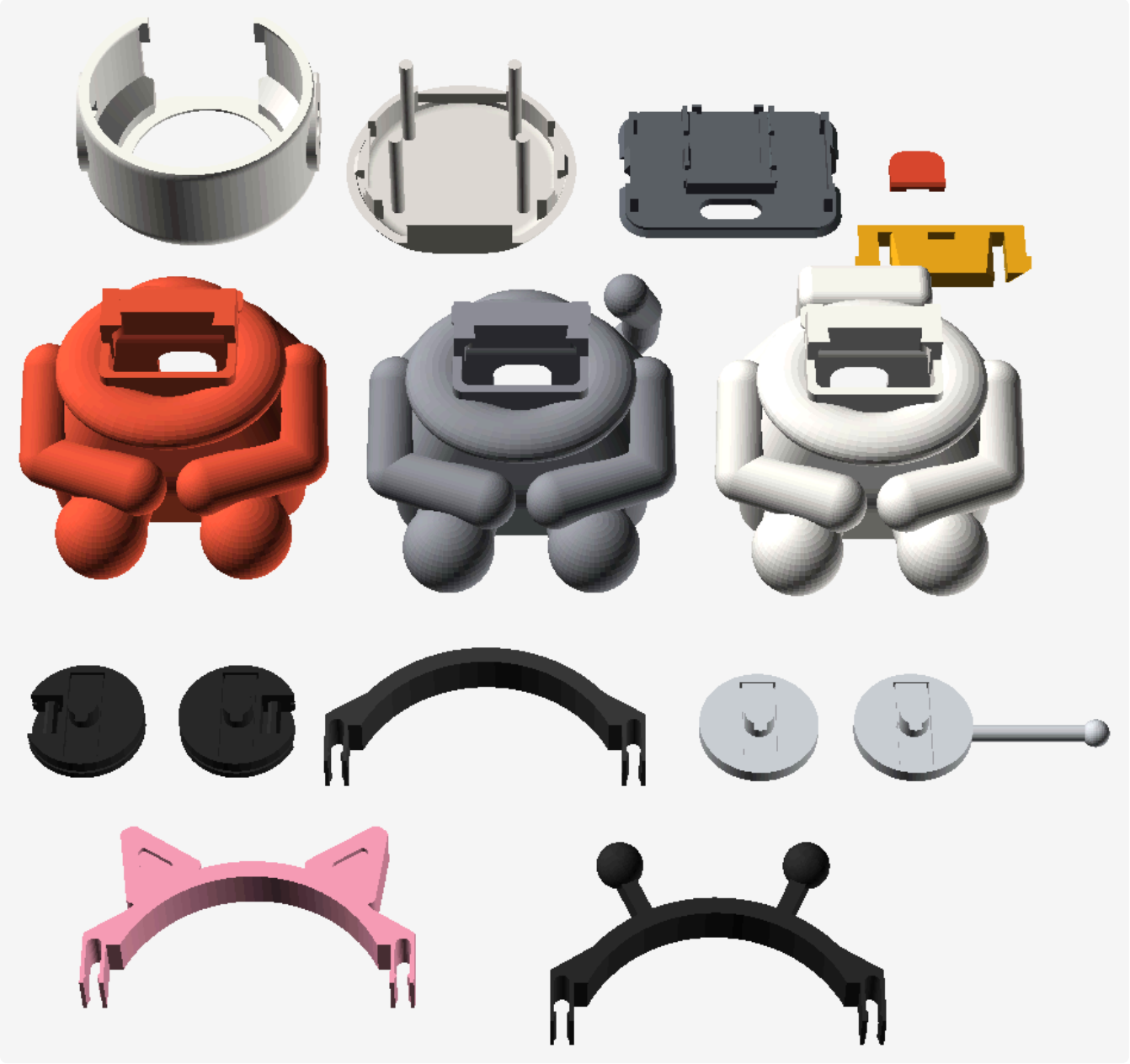
Hardware

Item	Qty	Notes
GC9A01 1.28″ round TFT module (38 mm round PCB, 7-pin)	1	The AITEXM-style module with onboard 3.3 V LDO
ESP32-S3 SuperMini	1	23.5 x 18 mm incl. USB-C
Silicone wire, 28–30 AWG, ~10 cm	7	Display to MCU
Heat-shrink, small	7	Insulate the solder joints on the display header

No screws and no glue are needed: every joint is a snap, a twist-lock or a key.

Printing

The STLs are already in print orientation. Load them as-is. Only the bodies need supports.





Part	Orientation (as exported)	Supports	Notes
head_shell	Face down	None	Use 0.12–0.16 mm layers for a cleaner bezel. The side keyholes bridge 3 mm
head_cap	Back face down	None	Four posts stand up; print slowly or enable minimum layer time
neck_key	Flat, as exported	None	Small and loaded: use 4 walls or 100% infill
cup_left / cup_right / cup_mic	Outer face down	None	Peg and lugs point up. The spring beam under the peg (20 mm) and the bar in the band slot (9 mm) print as bridges, so keep the part fan on. The mic boom lies flat on the bed

Part	Orientation (as exported)	Supports	Notes
plug / pod_left / pod_right	Outer face down	None	Print the plug twice. Same 20 mm spring-beam bridge. The antenna lies flat on the bed
All bands	Flat, back face down	None	Ears, horns, crown and plates are part of the flat outline, and so are the two clip legs
body_bare / body_hoodie / body_cat / body_astro	Upright	Tree, build plate only	Supports go under the arms, legs, collar, hood bag, tail or backpack. The two neck posts and the catch ribs inside need none
body_skeleton	Upright	Tree, build plate only	The thin arm bones need support along their length; remove it gently
base	Flat, cradle up	None	
usb_cover	Outer face down	None	

Suggested settings for PLA: 0.2 mm layers, 3 walls, 15% infill, no brim. One robot is about 70 cm^3 of solid model, roughly 40–55 g of filament.

Wiring

Seven wires run from the display header down the neck to the SuperMini. The GPIOs below are the S3's default FSPI pins, and all of them sit on the board's top pin row next to 5V and GND.

Display pin	SuperMini pin	Function
VCC	5V	Feeds the display's onboard 3.3 V regulator
GND	GND	
SCL	GPIO 12	SPI clock

Display pin	SuperMini pin	Function
SDA	GPIO 11	SPI MOSI (write-only, no MISO)
CS	GPIO 10	Chip select
DC	GPIO 9	Data / command
RST	GPIO 8	Reset

Keep the wires about 10 cm long. That is enough to solder the MCU outside the body, then fold the slack into the cavity. The backlight is hard-wired on, so no pin is needed for it.

Assembly

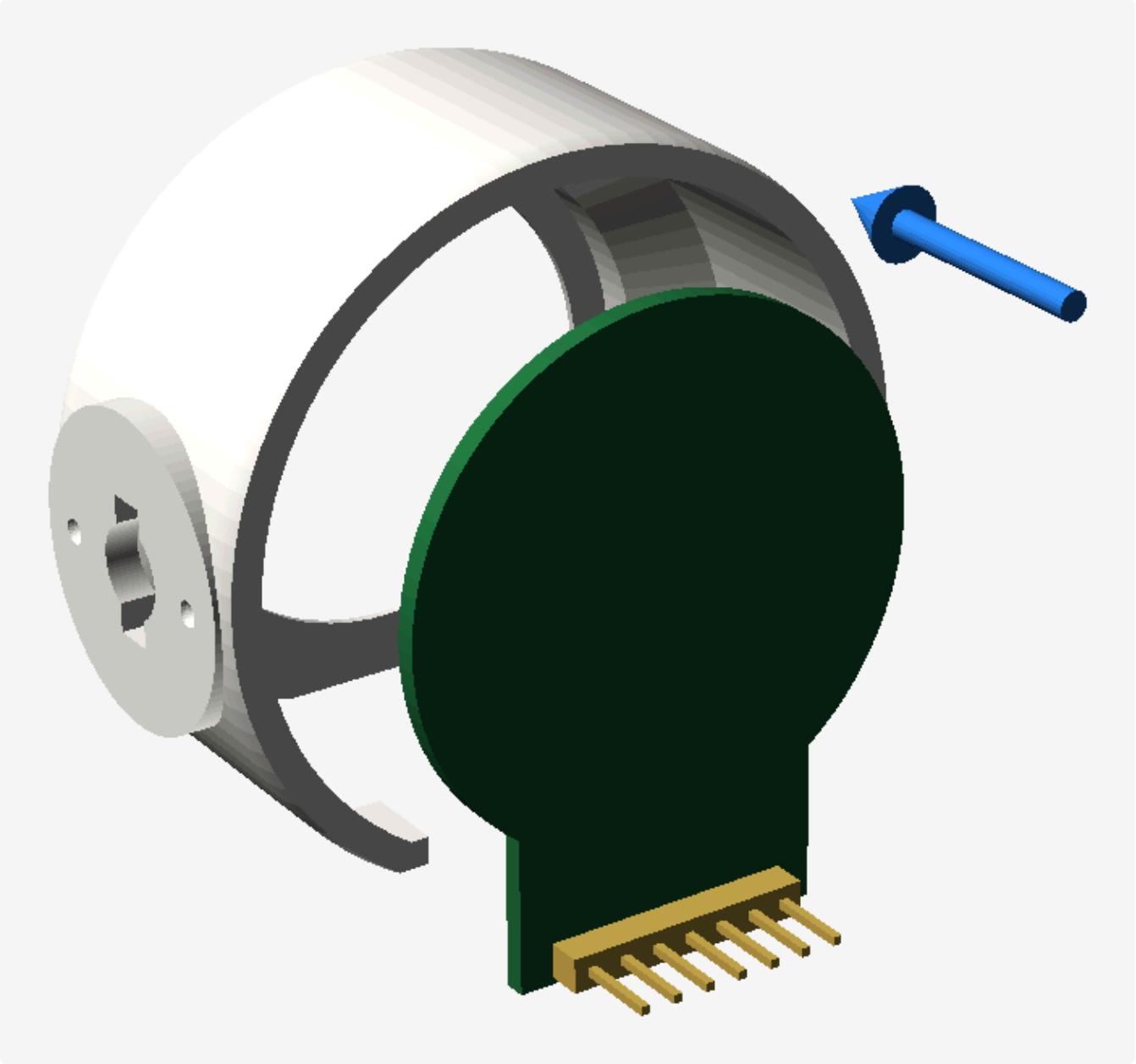
No glue is used anywhere. The order matters: the head goes onto the neck, then the wedge key, then the back cap, because the cap closes behind the key. Every motion below was collision-checked in CAD, so if a part binds, it is a print tolerance issue (see Fit tuning).

Step 1 — Wire the display

1. Solder the seven wires straight onto the display's header pins, close to the plastic spacer. Don't use Dupont plugs: they're too long for the neck.
2. Slide heat-shrink over each joint and let the wires hang straight down from the tab. Keep them at or below the pin row: the neck key later slides in just above it.
3. Check the back of the round PCB at the 45° diagonals, 17 mm from centre. The four cap posts press there. If a tall component sits under one, snip that post shorter.

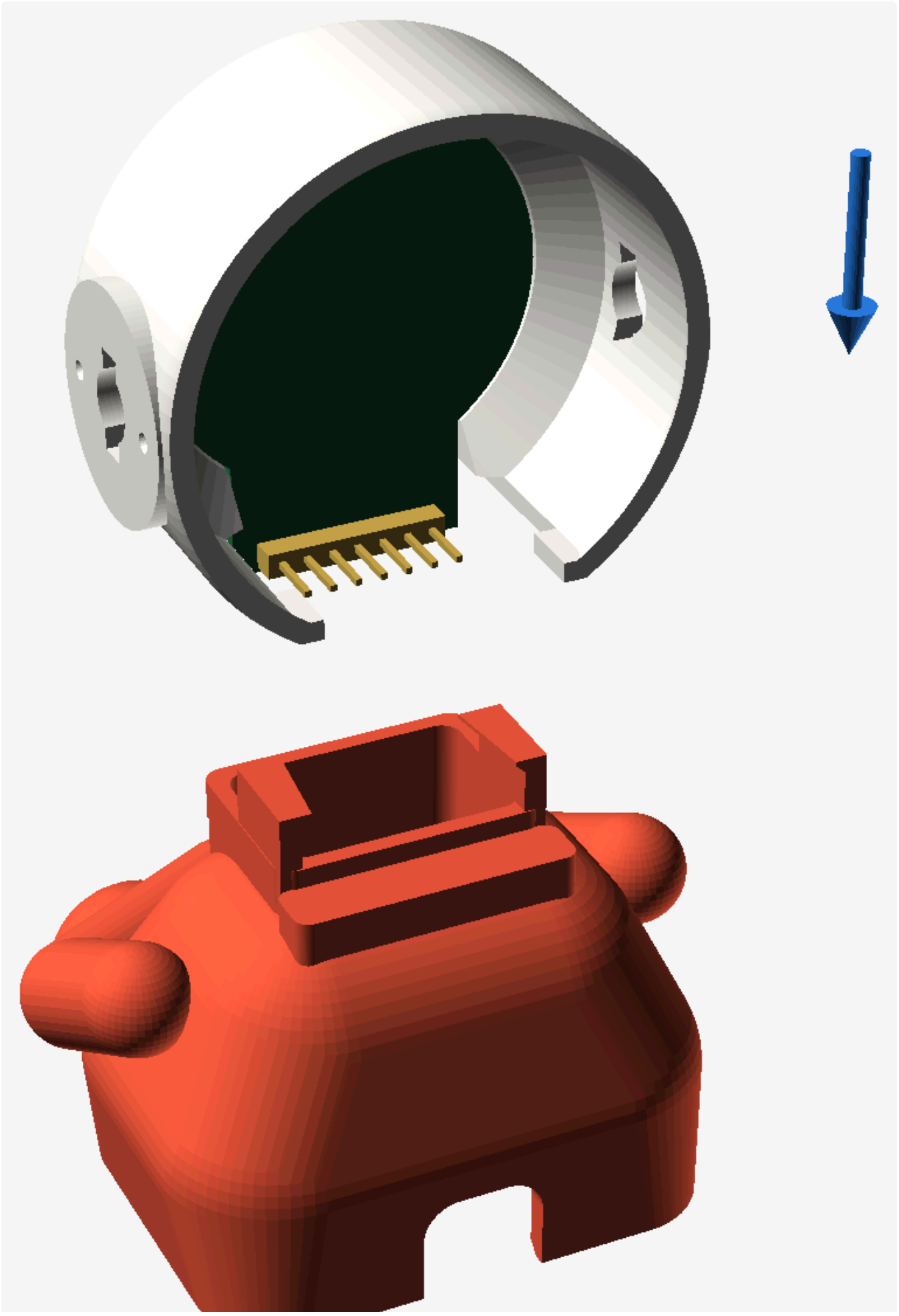
Step 2 — Slide the display into the head

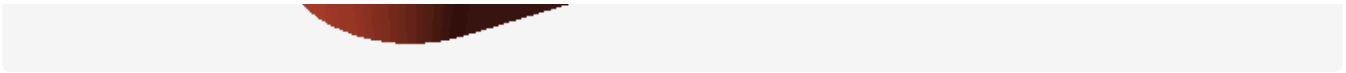
1. Hold the head shell face down. Feed the display in from the open back, glass first.
2. The tab and its wires drop through the slot in the head's floor as the display slides forward.
3. Push until the glass sits flat against the bezel lip, with the tab pointing straight down.



Step 3 — Seat the head on the neck

1. Feed the seven wires down through the square neck opening until they come out of the open bottom of the body.
2. Lower the head shell, with the display inside and the cap still off, onto the neck.
3. The two posts on top of the neck come up through the socket in the head's floor, one on each side, behind the display. The head now rests on the neck's top at the front and on the flat rest block behind the neck. It still lifts off freely: the key in the next step is what locks it.

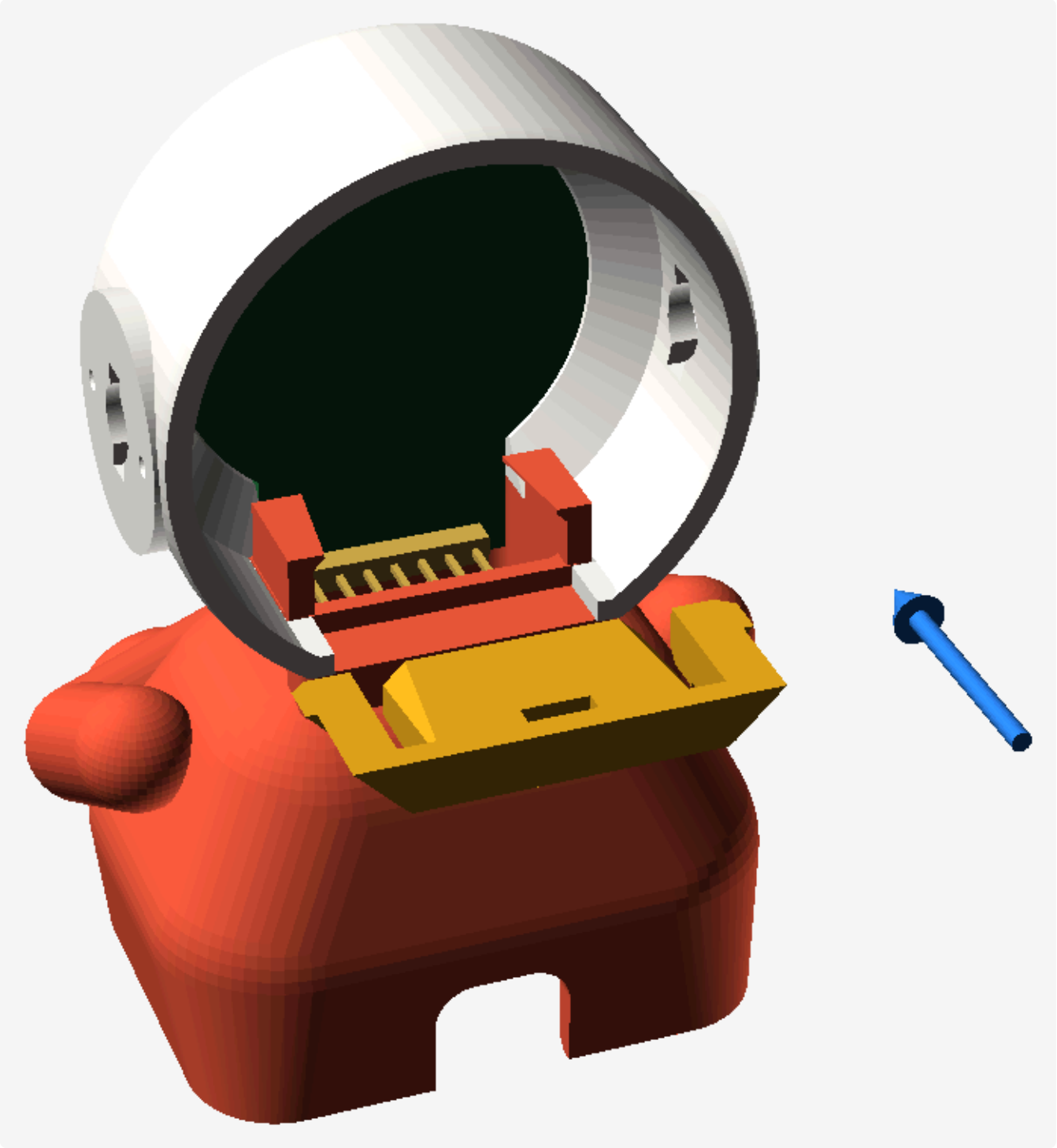


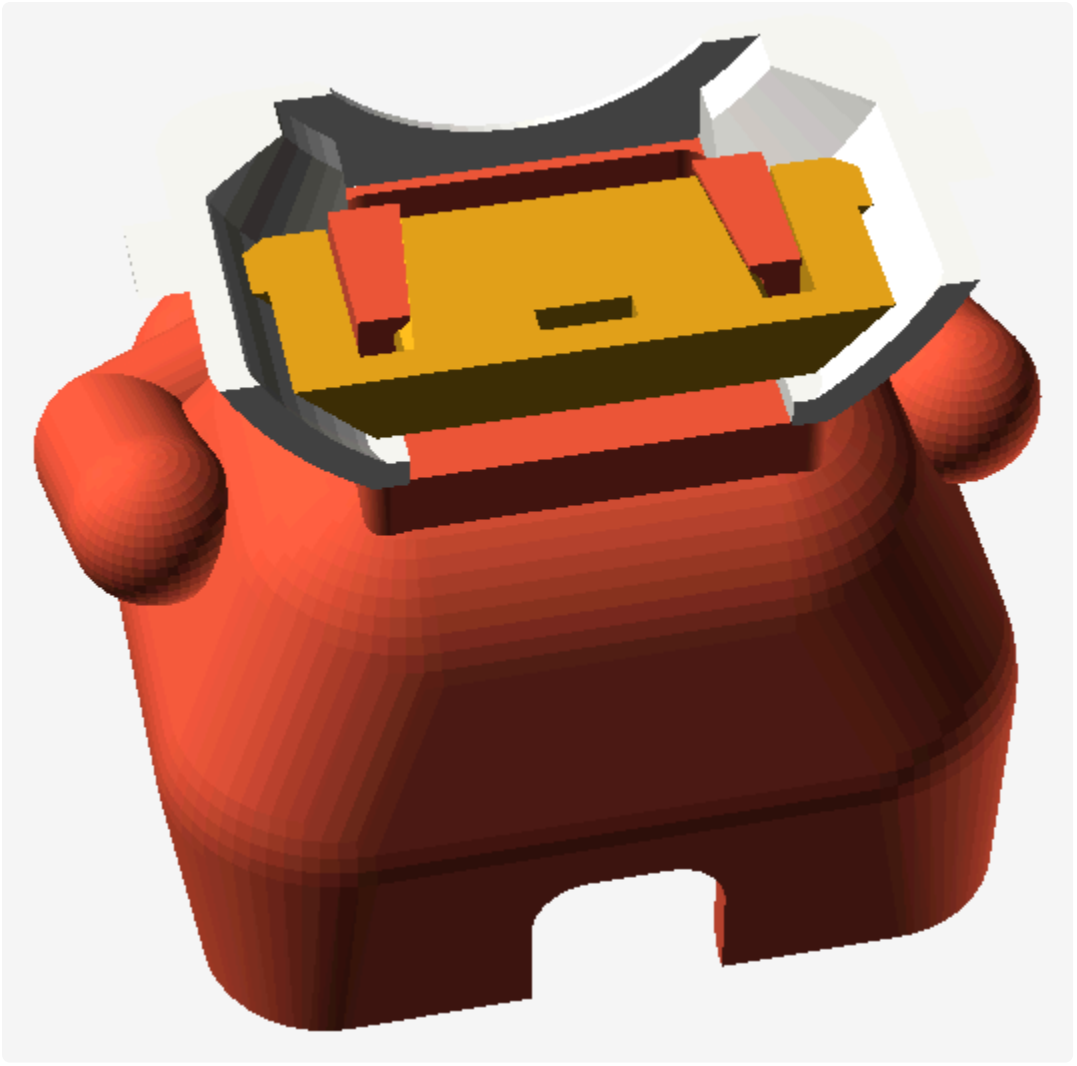


Step 4 — Lock the head with the wedge key

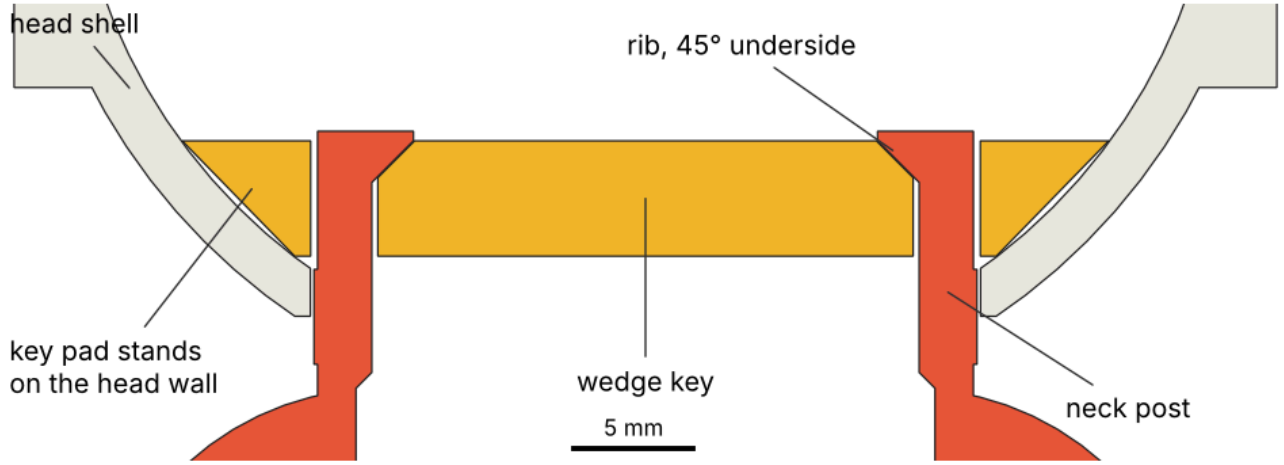
1. Hold the key flat, forked end first, and feed it into the open back of the head along the floor.
2. The wide middle tongue goes between the two neck posts, and the two outer prongs pass outside them. Let the forked end ride over the rim of the opening: it has to sit about 1 mm high there, then it drops down inside.
3. Push the key forward with a fingertip until it stops firmly. Its bevelled edges run under the ribs on the posts. The ribs get deeper towards the front, so the last millimetres pull the neck up into the head and clamp the head onto its rests. The head should now have no play at all.
4. To release it later, hook a small screwdriver into the slot in the key and pull it straight back.

The key normally stops with its back edge 0–4 mm inside the rim of the opening. If it goes in until it touches the posts and the head is still loose, or it goes tight while its back edge is still outside the rim, see Fit tuning.

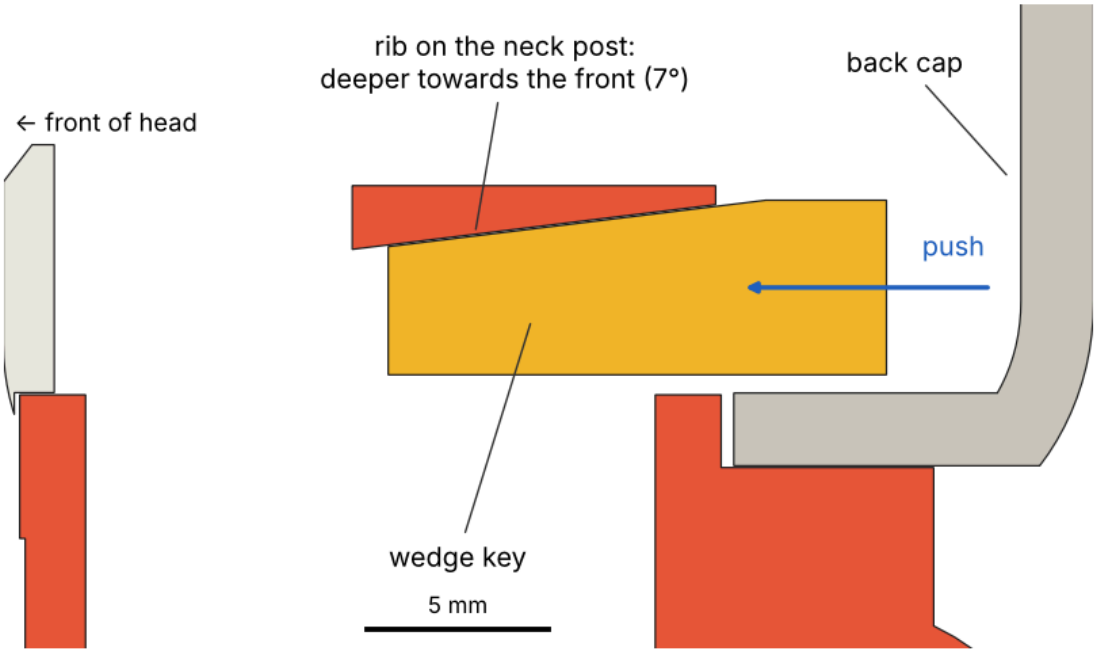




Neck lock, section seen from behind

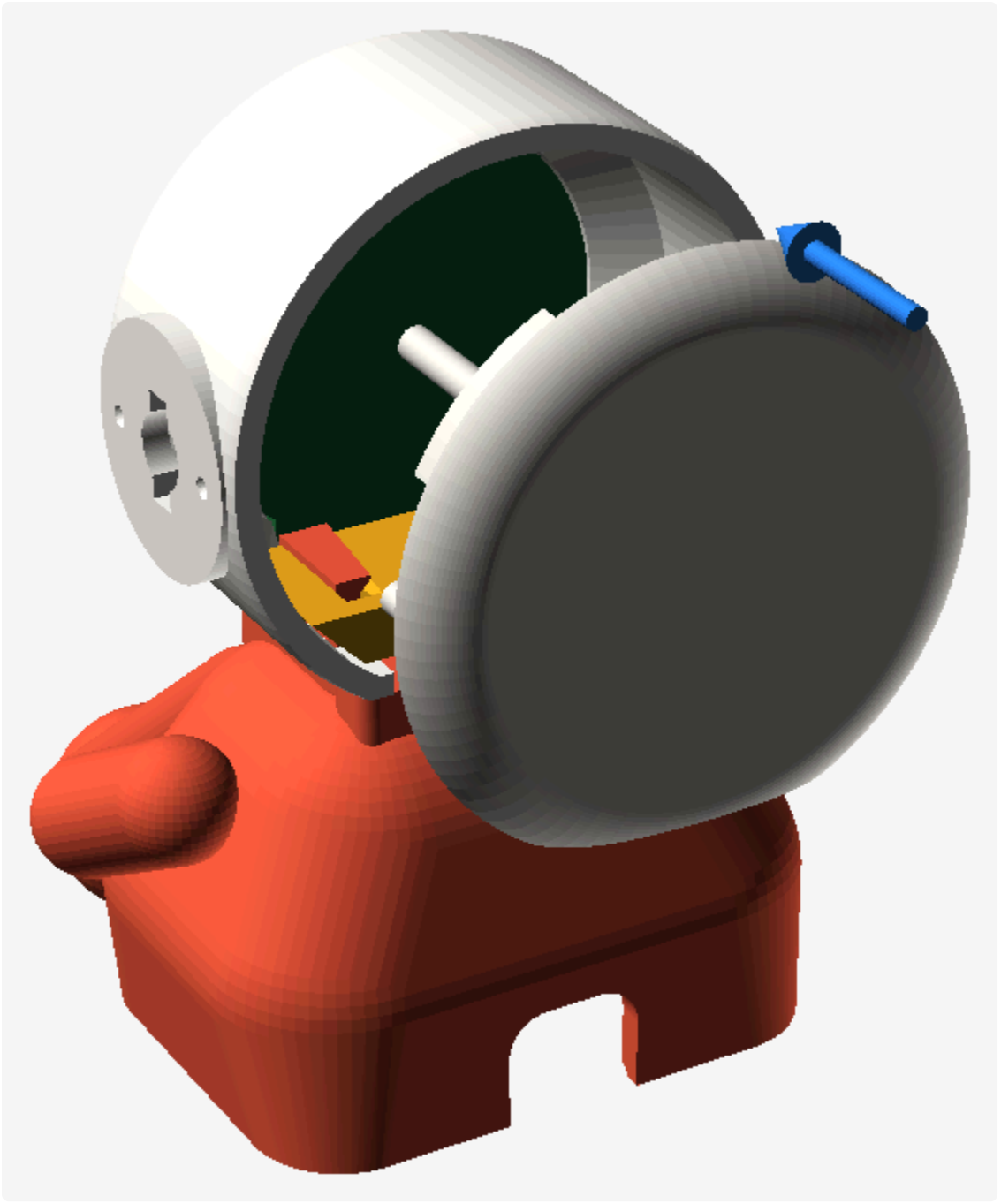


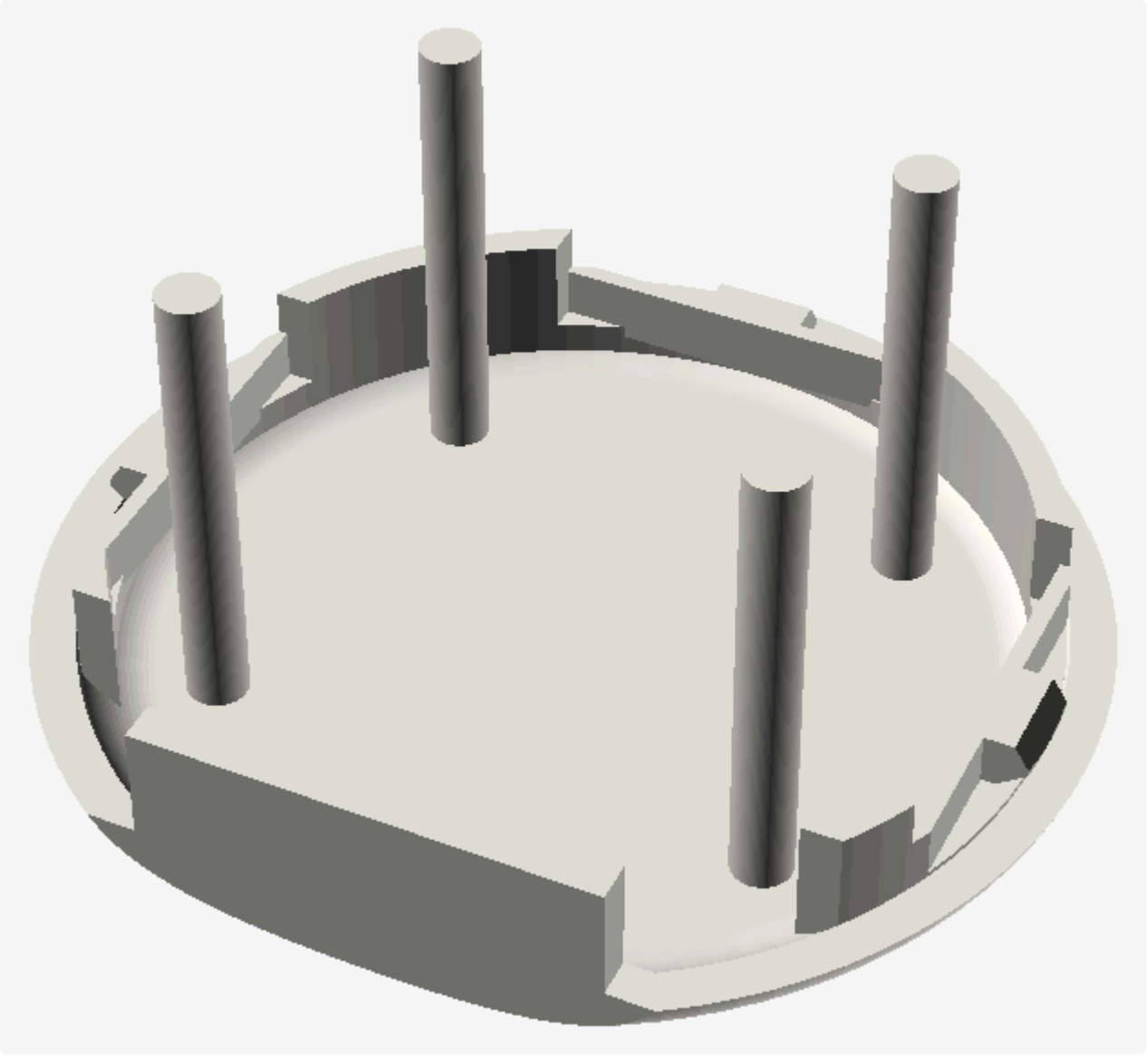
Side section through a rib: pushing the key in pulls the neck up



Step 5 — Close the back cap

1. Press the back cap straight in from behind until the three teeth on its rim click in behind the lip of the shell opening.
2. The four posts clamp the display against the bezel. A block at the bottom of the cap sits right behind the key, so the key cannot work its way out while the cap is on.
3. To take the head off later: pull the cap firmly straight back, draw the key out, then lift the head.



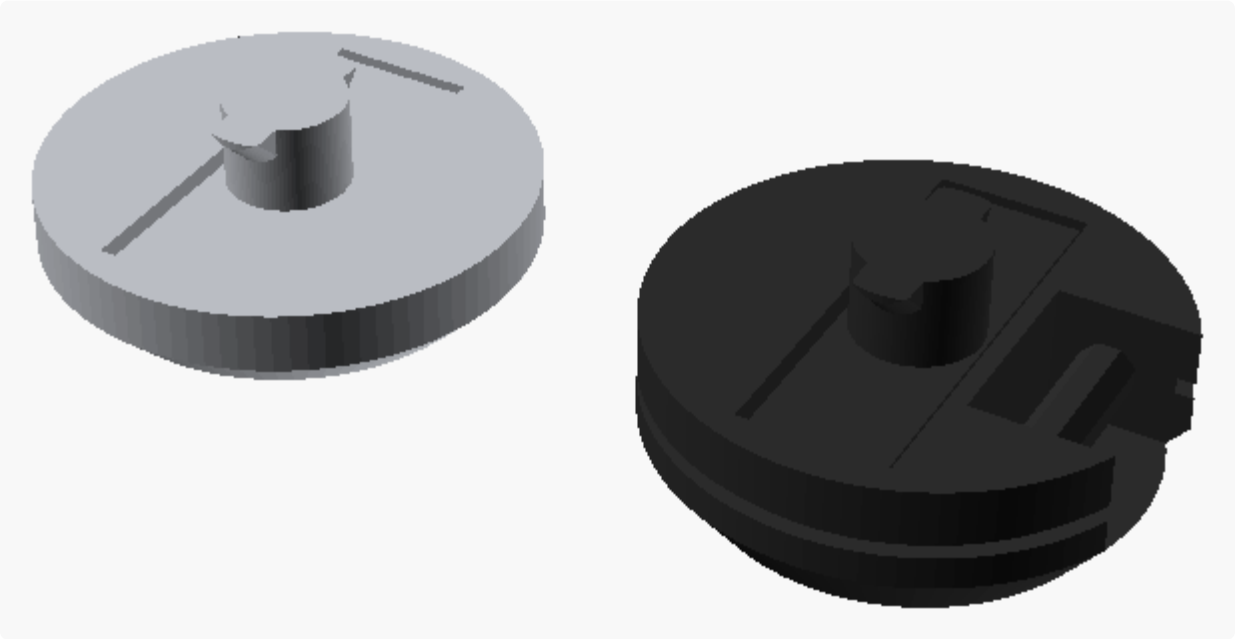


Step 6 — Twist-lock the side pieces, snap on the band

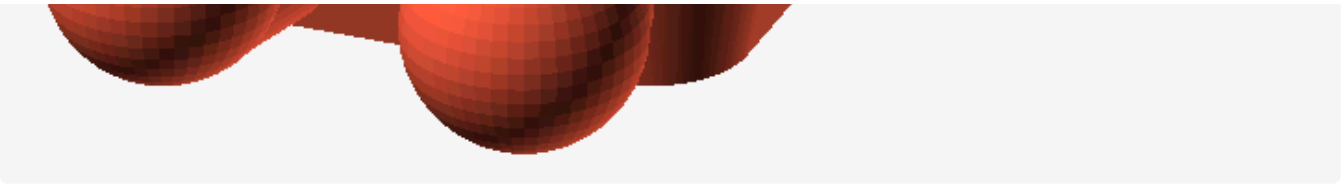
1. Every side piece has a small dot on its outer face. Hold the piece so the dot points up, and push its peg into the keyhole on the side of the head until the piece sits flat against the head.
2. Twist it a quarter turn so the dot moves to the back: a cup until its band slot faces up, a pod until its antenna points up. The twist is firm from the start. The peg stands on a spring beam, and turning it winds the two lugs in behind the wall, so the piece ends up pulled tight against the head.
3. Do the same on the other side. Then hold the band over the head, set its two legs into the cup slots and press down on both ends until they click. Each leg is a U-shaped clip that snaps over a bar inside the cup.
4. The band now also stops the cups from turning back. To remove it, pull both ends straight up; it lets go with a firm tug.

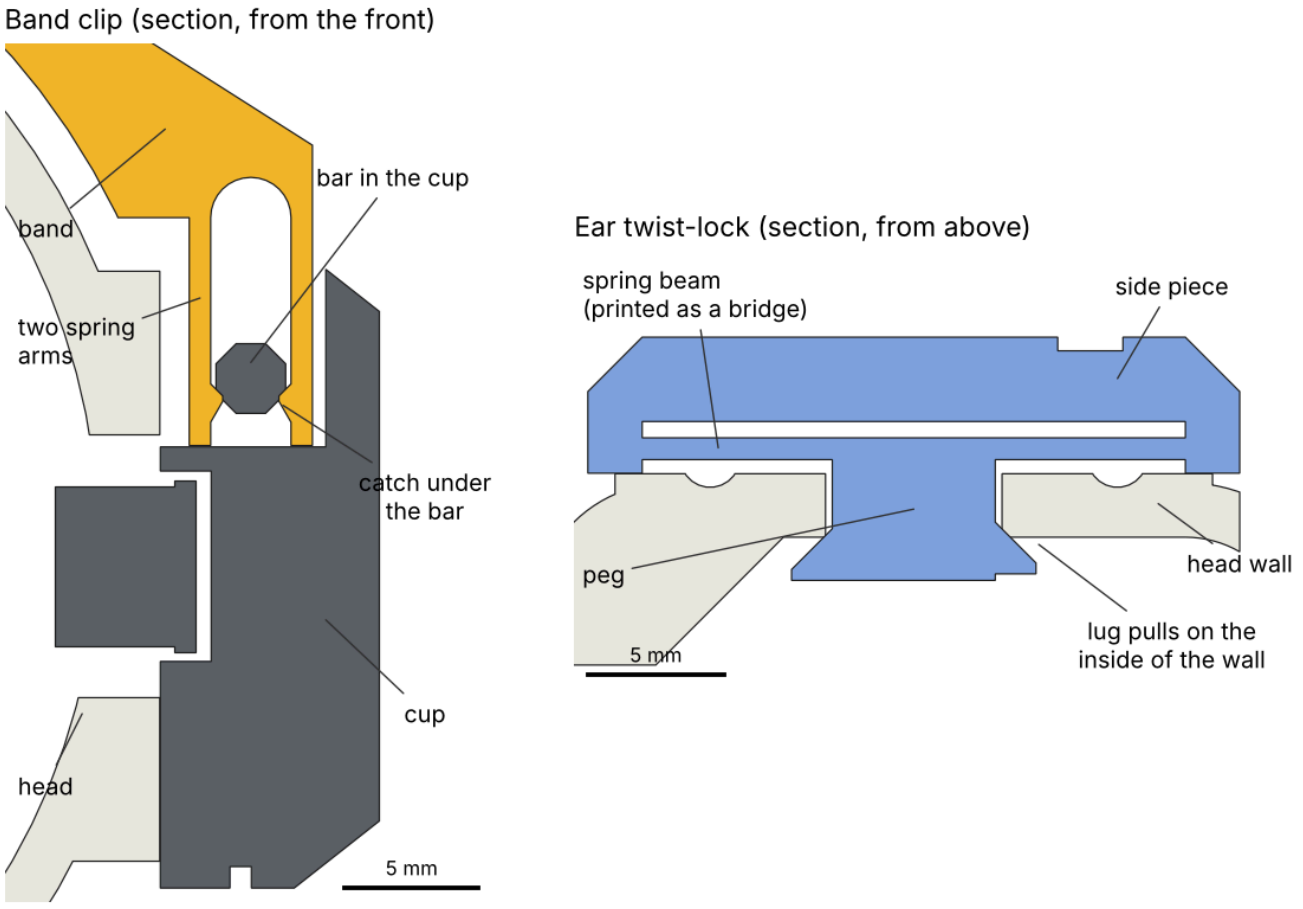
5. Plugs (Template, Skeleton) have no fixed angle: dot up, push in, quarter turn either way.
6. Gamer theme: the mic cup goes on the robot's left.

The dot matters because the head's inner wall is curved. The lug on the dot side is the tight one and must travel through the upper half of the keyhole; the other lug is cut 0.24 mm deeper to clear the lower half.



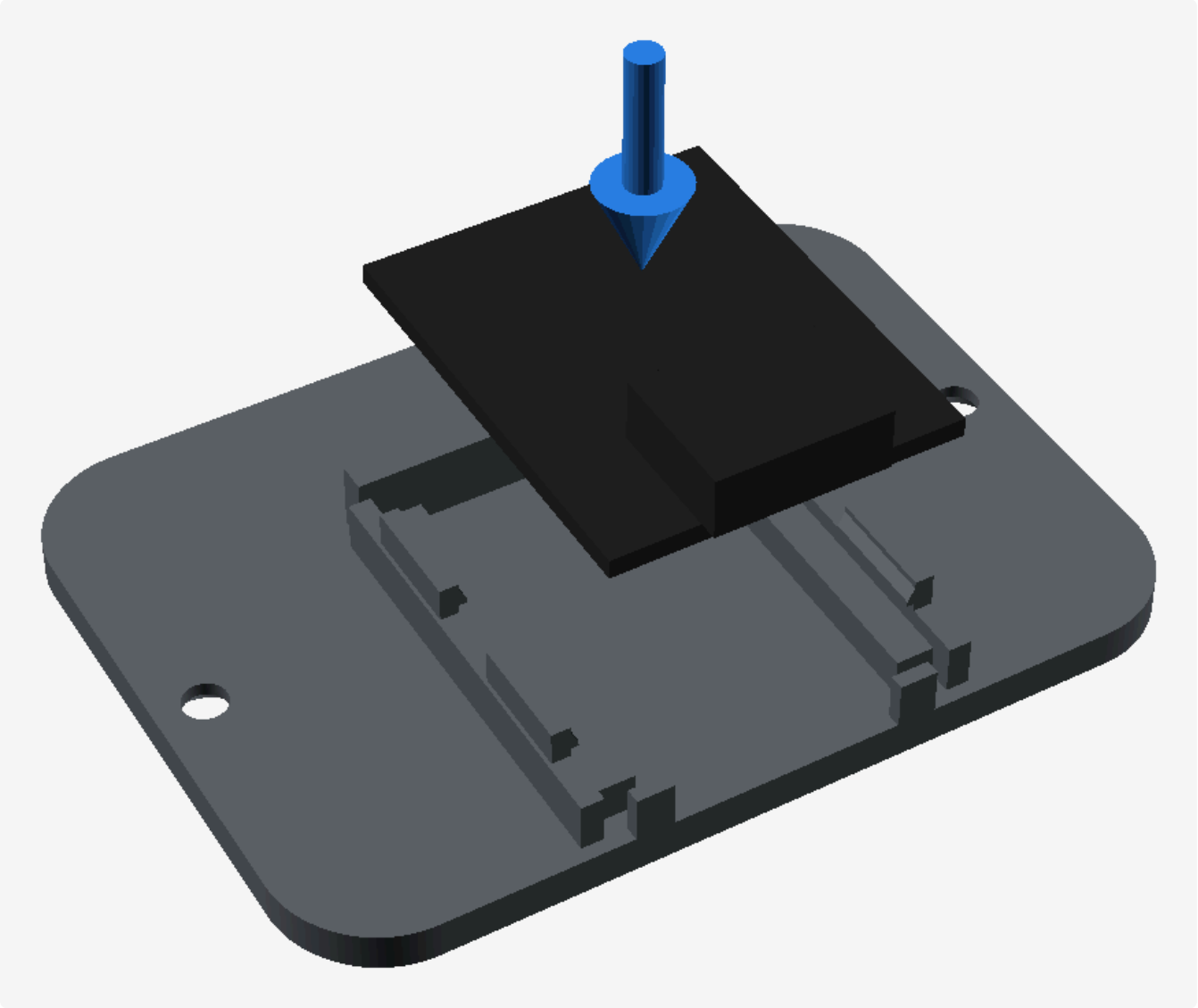






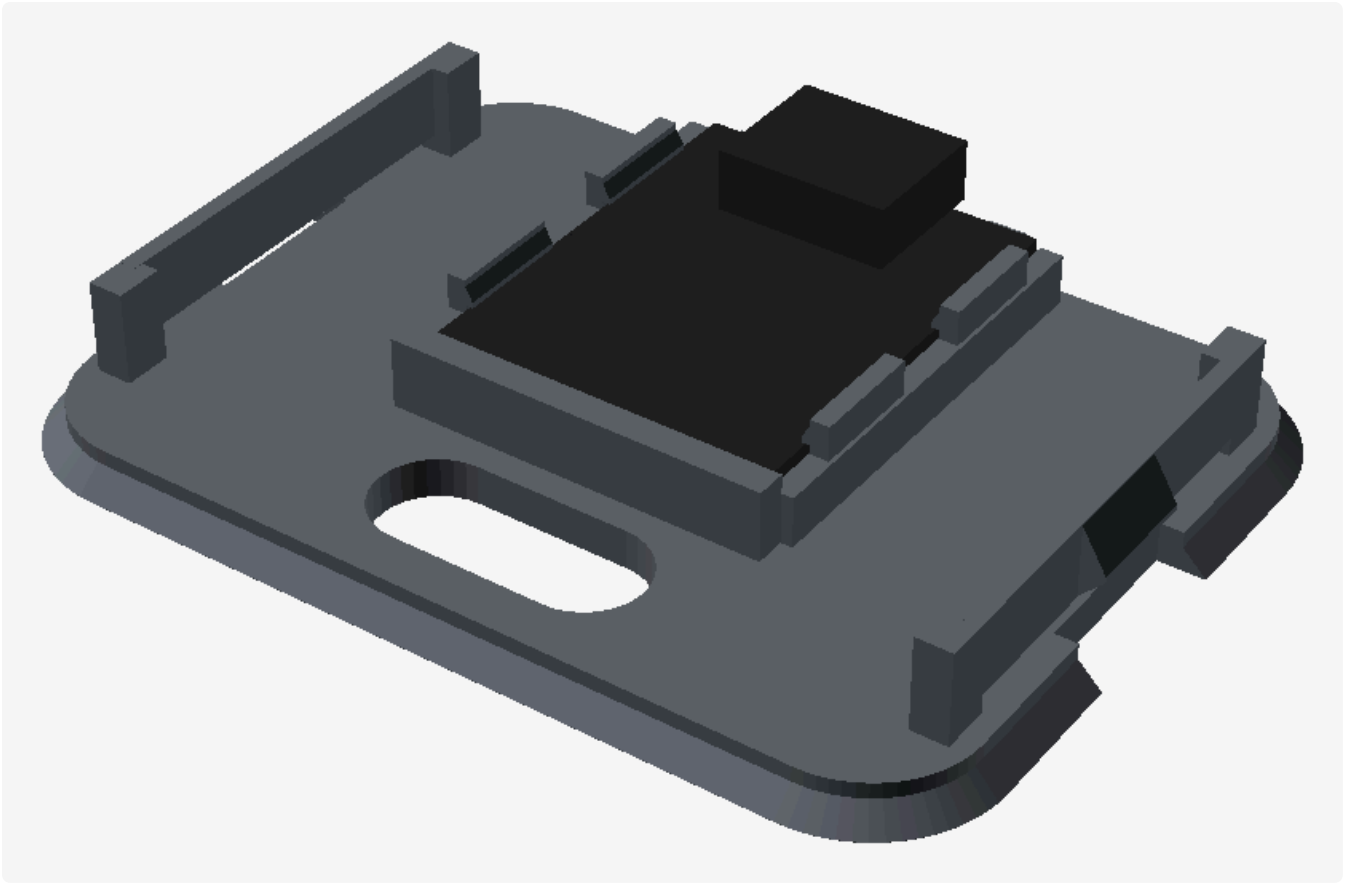
Step 7 — Mount the SuperMini

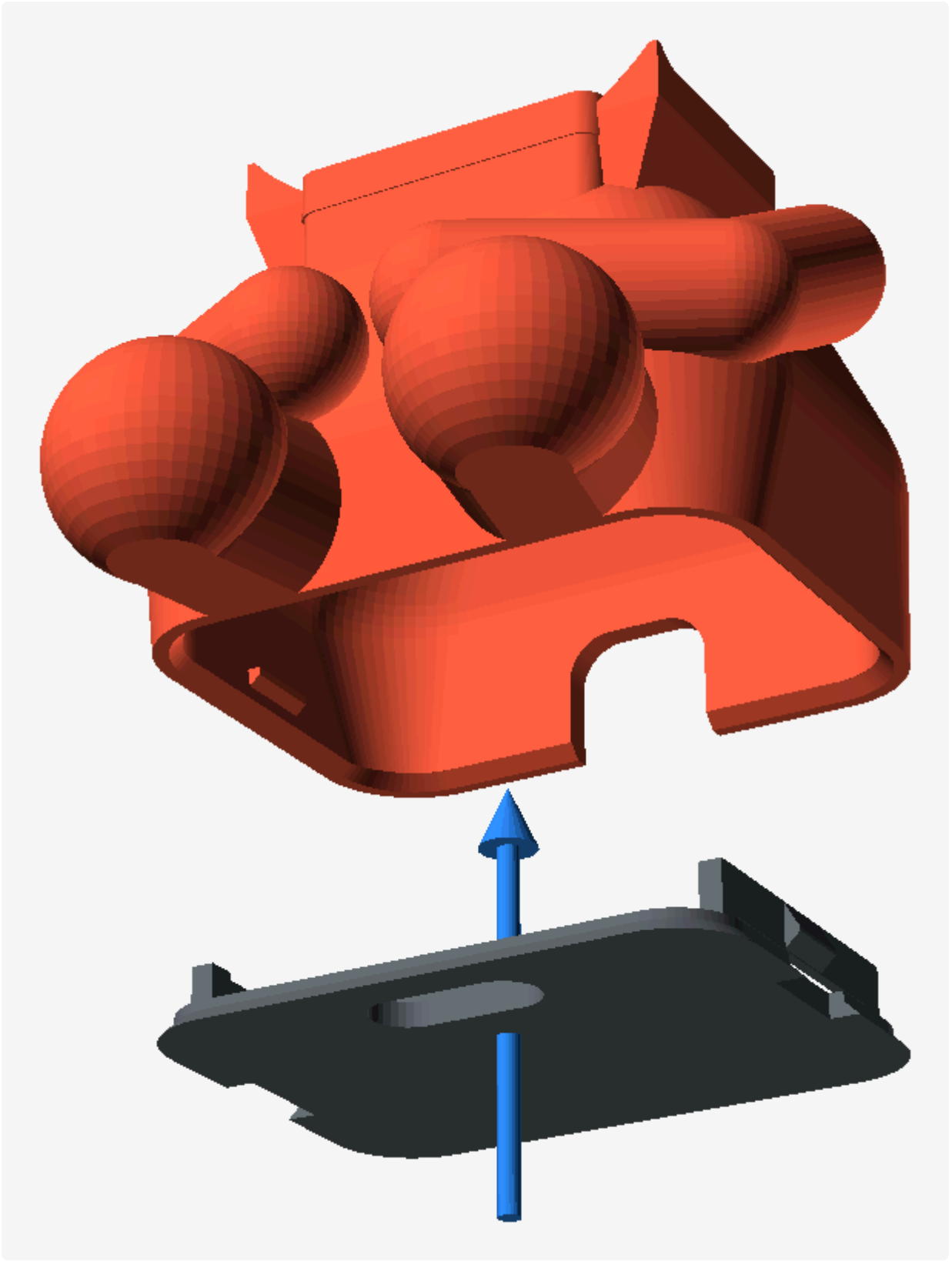
1. Solder the wires to the SuperMini (see Wiring), from the top side of the board.
2. Press the board into the base plate's cradle, components up, USB-C toward the open end. The four snap tabs click over the board edges.
3. If your board is thicker than 1.0 mm and the tabs won't latch, set `mcu_t` to your board's thickness and reprint the base.



Step 8 — Close the base

1. Fold the wire slack into the body. Push the base plate up from below, with the USB-C end toward the back of the robot.
2. The receptacle slides up through the open-bottom notch in the back wall. The plate's sloped edge seats in the matching cone in the body rim, flush with the bottom.
3. Press until the two side teeth click over the ribs inside the body. To open it again, hook a fingertip or a small tool in the pull slot and pull the plate straight down.





Step 9 — Finish

1. Plug a USB-C cable in through the notch on the lower back and flash the test sketch below.
2. Press the USB cover into the notch whenever no cable is connected. A fingernail slot on its bottom edge lets you pry it out.

3. To change theme, pull the band straight up off the cups, twist the side pieces back a quarter turn and pull them off, then fit another set.



First power-up

The sketch below draws two happy amber eyes and blinks every few seconds. If you see them, the wiring is right. It uses the Arduino_GFX library (moononournation), installed from the Arduino Library Manager. Select board "ESP32S3 Dev Module" with "USB CDC On Boot: Enabled".

```
#include <Arduino_GFX_Library.h>

// DC, CS, SCK, MOSI, MISO
Arduino_DataBus *bus = new Arduino_ESP32SPI(9, 10, 12, 11, GFX_NOT_DEFINED);
// RST, rotation, IPS panel
Arduino_GFX *gfx = new Arduino_GC9A01(bus, 8, 0, true);

const uint16_t AMBER = 0xFD20, BG = 0x0000;

void eyes(bool open) {
  gfx->fillRect(40, 70, 160, 60, BG);
  for (int x : {82, 158}) {
    if (open) gfx->fillArc(x, 120, 32, 22, 180, 360, AMBER);  // ^ ^
    else      gfx->fillRect(x - 32, 112, 64, 8, AMBER);       // - -
  }
}

void setup() {
  gfx->begin(40000000);  // 40 MHz SPI
  gfx->fillScreen(BG);
  eyes(true);
}

void loop() {
  delay(random(2500, 5000));
  eyes(false); delay(120);
  eyes(true);
}
```

If the screen stays black, check RST and CS first. If the image is mirrored or the colours are inverted, change the rotation or the IPS flag in the constructor.

Fit tuning

All fits come from parameters at the top of round_bot.scad. Change one, set `part` to the piece you need, and export again with F6 then F7 in OpenSCAD.

Symptom	Parameter	Default	Change
Back cap too tight / loose	`spigot_clr`	0.15 mm	Raise / lower in 0.05 steps
Display rattles, or cap won't seat	`post_gap`	0.2 mm	Lower to clamp harder / raise to relieve
Display doesn't drop into the pocket	`clr`	0.3 mm	Raise to 0.4 (also loosens neck and plate)
Black circle off-centre in the bezel	`screen_off_z`	0 mm	+ moves the window up
Side piece still has play when locked	`ear_tension`	0.15 mm	Raise to 0.25
Side piece is too stiff to twist, or its spring beam cracks	`ear_tension`	0.15 mm	Lower to 0.05, or set `beam_t` to 0.6 for a softer spring
Side piece jams as the twist starts, dot side up	`lug_relief`	0.24 mm	Raise to 0.35
Band is too hard to press on / pulls off too easily	`band_bite`	0.25 mm	Lower to 0.15 / raise to 0.35
Band rattles on the cups	`band_pre`	0.1 mm	Raise to 0.2
SuperMini won't latch / is loose	`mcu_t`	1.0 mm	Set to your measured PCB thickness
Cable plug won't reach the port	`usb_w`, `usb_h`	12.5 x 7.5 mm	Widen for chunky cable overmolds

If the head shell still slides off the cap, raise `snap_bite` from 0.42 to 0.55 mm and reprint the cap. If the cap is too hard to push in, lower it to 0.3 mm.

Head and base fit. The first two rows need only the small key reprinted:

Symptom	Parameter	Default	Change
Key goes in until it touches the posts and the head is still loose	`key_fat`	0 mm	Set to 0.15. Each 0.1 makes the key go tight 0.8 mm earlier
Key goes tight too early: its back edge stays outside the rim and the cap won't close	`key_fat`	0 mm	Set to -0.15

Symptom	Parameter	Default	Change
Head rocks front to back even with the key tight	`neck_plug`	1.95 mm	Raise to 2.05 so the neck top reaches the socket ceiling (reprint the body)
Neck plug too tight / loose in the head socket	`neck_fit`	0.15 mm	Lower / raise in 0.05 steps (reprint the body)
Base plate drops out / too hard to click in	`base_bite`	0.5 mm	Raise to 0.65 / lower to 0.35
Base plate rattles sideways / won't enter	`base_clr`	0.12 mm	Lower to 0.08 / raise to 0.2
Base plate moves up and down when clicked in	`base_play`	0.15 mm	Lower to 0.05

Make your own theme

Start from the template and change only the accessories; the core never needs reprinting. In round_bot.scad, copy one of these modules and edit its outline.

To add	Start from	Interface to keep
A side piece (ears, horns, bolts)	`plug()`	Keep `ear_lock()`: the peg, its two lugs and the spring beam. Cut `ear_cut()` from your shape so the beam is free to move. Stay outside 26.3 mm from the head axis so it can twist
A headband topper	`topper_2d()`	Draw a flat outline above the band (radius 31 mm). It is extruded to the 9 mm band width and prints flat
A body	`body("bare")`	Add shapes to the union. Keep the neck, `neck_posts()`, the rest block, the cavity and the USB notch

Every band needs the two cups, because the cups hold its ends.